

Curriculum Vitae

Raffael Jonas

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Education/Academic Background

Master in Astronomy University of Vienna

October 2025 – September 2027

Bachelor in Astronomy University of Vienna

October 2022 – September 2025

Average grade: 1.74 (with 1 being the best and 5 the worst)

Higher Technical Institute (HTL) for Computer Science and Engineering Spengergasse Vienna.

2015 – 2020

Research Experiences

Bachelor's Thesis: Excursion Set Theory Halo Mass Function

Data Science in Astrophysics & Cosmology Group, University of Vienna

Supervisor: MSc Agata Wiśłocka

31/1/2025 – 05/9/2025

Constructed halo mass functions within Excursion Set Theory using four smoothing filters: Sharp-k, Top-hat, Gaussian, and Smooth-k and comparing the results to analytical models and tools like Colossus & Mira-Titan. Reconstructing the methodology of Wiśłocka et al., "Excursion sets with a 'perfect' collapse model" (<https://academic.oup.com/mnras/article/541/2/880/8185403>) using the Smooth-k filter.

Paid Research project: Investigating Galaxy Bias with the TNG300 Simulation

Data Science in Astrophysics & Cosmology Group, University of Vienna

Project lead: Prof. Oliver Hahn

01/10/2025 – 08/06/2026

Performing large-scale particle data analysis on the TNG300 hydrodynamical simulation (TBs) using the Vienna Scientific Cluster (VSC). Identifying and locating most bound particles and finding its location in a other snapshot. Analyzing spatial clustering, galaxy bias, and secondary galaxy properties clustering. Collaborating in a research team with version control, code documentation, and presenting results in group meetings.

Non-research Experiences/Work/Volunteering Activities

Project: Design and Implementation of a Player Statistics Tracking Tool for Spikeball/Roundnet

- Inferring detailed statistics based on input order, allowing the game to be tracked simply by recording ball possession sequences.
- Statistical analysis spanning multiple games and tournaments.

Tutor for Mathematics

2019 – now

- All school levels, including primary and secondary education.
- Experience in technical mathematics and economical mathematics.
- Preparation for university-level mathematics in natural sciences and business.

Athlete for the Austrian national team (Spikeball/Roundnet)

- World Championship 2022 (4th Place).
- European Championship 2023 (2nd Place).

Kitesurf Instructor

2016 – 2022

- Teaching kitesurfing to people of all ages, covering both the motor skills required for the sport and its theoretical foundations.

Skills

Programming Languages:

- C++ (advanced).
- Python (advanced, including Astropy, healpy, CAMB., analysis of TB worth of data).
- Java (advanced).
- Fortran (basics).
- C# (basics).
- SQL/MySQL (advanced).
- NoSQL (basics).

High-Performance Computing & Automation:

- Bash(advanced).
- Git (advanced).
- Slurm(basics).
- Large-scale astronomical data handling (HDF5, FITS)

Data Analysis & Machine Learning:

- Markov chain methods and stochastic modeling.
- Keras.
- Random forest.
- Gaussian Processes.

Languages

- German C2 (native).
- English C1.